Lab Experiment-9

Aim

To implement a rule-based Part-of-Speech (POS) tagging system using regular expressions in Python to identify the grammatical categories of words in a sentence.

Algorithm

1. Start the program.
2. Import the re module for regular expression operations.
3. Read a sentence from the user.
4. Tokenize the sentence into individual words using regular expressions.
5. Define regular expression rules for determiners, pronouns, verbs, adverbs, adjectives, nouns, and punctuation.
6. Match each word against the rules and assign the corresponding POS tag.
7. Assign a default noun tag (NN) if no other rule matches.
8. Display each word with its POS tag.
9. Stop the program.

Code:

```
import re

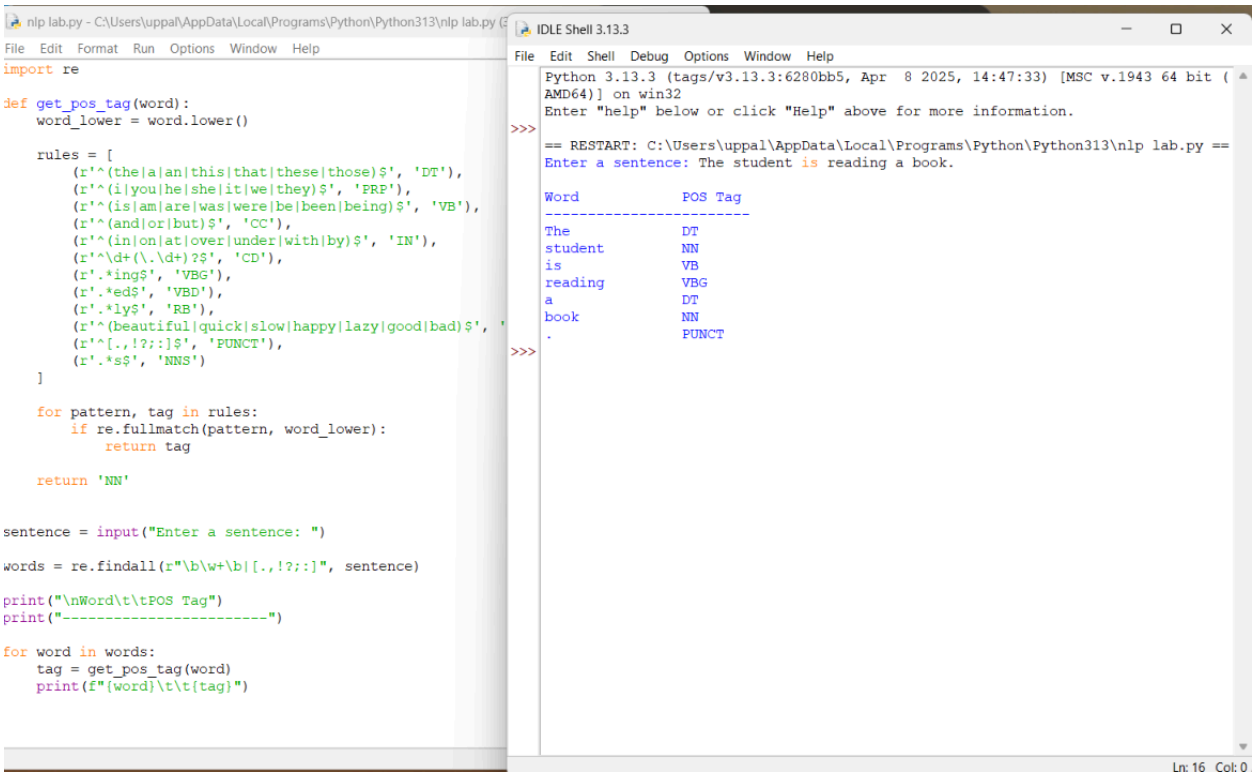
def get_pos_tag(word):
    word_lower = word.lower()
    rules = [
        (r'^(the|a|an|this|that|these|those)$', 'DT'),
        (r'^(i|you|he|she|it|we|they)$', 'PRP'),
```

```
        (r'^(is|am|are|was|were|be|been|being)$', 'VB'),
        (r'^(and|or|but)$', 'CC'),
        (r'^(in|on|at|over|under|with|by)$', 'IN'),
        (r'^\d+(\.\d+)?$', 'CD'),
        (r'.*ing$', 'VBG'),
        (r'.*ed$', 'VBD'),
        (r'.*ly$', 'RB'),
        (r'^(beautiful|quick|slow|happy|lazy|good|bad)$', 'JJ'),
        (r'^[.,!?;:]$', 'PUNCT'),
        (r'.*s$', 'NNS')
    ]
    for pattern, tag in rules:
        if re.fullmatch(pattern, word_lower):
            return tag
    return 'NN'
sentence = input("Enter a sentence: ")
words = re.findall(r"\b\w+\b|[.,!?;:]", sentence)
print("\nWord\t\tPOS Tag")
print("------------------------")
for word in words:
```

```
    tag = get_pos_tag(word)

    print(f"{word}\t\t{tag}")
```

Output:



```
import re

def get_pos_tag(word):
    word_lower = word.lower()

    rules = [
        (r'^(the|a|an|this|that|these|those)$', 'DT'),
        (r'^(i|you|he|she|it|we|they)$', 'PRP'),
        (r'^(is|am|are|was|were|be|been|being)$', 'VB'),
        (r'^(and|or|but)$', 'CC'),
        (r'^(in|on|at|over|under|with|by)$', 'IN'),
        (r'^\d+(\.\d+)?$', 'CD'),
        (r'.*ing$', 'VBG'),
        (r'.*ed$', 'VBD'),
        (r'.*ly$', 'RB'),
        (r'^(beautiful|quick|slow|happy|lazy|good|bad)$',
        (r'^[.,!?;:]$', 'PUNCT'),
        (r'.*s$', 'NNS')
    ]

    for pattern, tag in rules:
        if re.fullmatch(pattern, word_lower):
            return tag

    return 'NN'


sentence = input("Enter a sentence: ")

words = re.findall(r"\b\w+\b|[.,!?;:]", sentence)

print("\nWord\t\tPOS Tag")
print("-------------------------")

for word in words:
    tag = get_pos_tag(word)
    print(f"{word}\t\t{tag}")
```

```
Python 3.13.3 (tags/v3.13.3:6280bb5, Apr  8 2025, 14:47:33) [MSC v.1943 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
== RESTART: C:\Users\uppal\AppData\Local\Programs\Python\Python313\nlp lab.py ==
Enter a sentence: The student is reading a book.

Word            POS Tag
-------------------------
The             DT
student         NN
is              VB
reading         VBG
a               DT
book            NN
.               PUNCT
>>>
```

Result

Thus, a rule-based Part-of-Speech tagging system was implemented successfully using Python regular expressions. The program identifies words in a sentence and assigns POS tags according to predefined grammatical rules without requiring external language datasets.